



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

P2 Electricity: Test Review Checklist

Test: Wednesday 7 October, Period 4

Everything on the test is on this page. Tick a box when you can do it without looking at your notes.

1. Circuit diagrams

- ☐ Draw and recognise the symbols for a switch (open and closed), cell, battery, resistor, variable resistor, lamp, fuse, diode, LED, ammeter, voltmeter, thermistor and LDR.
- ☐ Draw and read circuit diagrams.
- ☐ Connect an ammeter in **series** and a voltmeter in **parallel**.

2. Current, potential difference (voltage), resistance

- ☐ A circuit needs a source of potential difference (voltage) for charge to flow.
- ☐ Current is the rate of flow of charge: $Q = It$ (C, A, s).
- ☐ The current is the same at every point in a single loop.
- ☐ For the same p.d., more resistance means less current.
- ☐ Use $V = IR$ (V, A, Ω), and rearrange it.

3. Resistors and I–V graphs

- ☐ Ohmic conductor: at constant temperature, current is directly proportional to p.d.; resistance stays constant.
- ☐ Filament lamp: resistance increases as the filament gets hotter.
- ☐ Diode: current flows one way only; very high resistance in reverse.
- ☐ Thermistor: resistance decreases as temperature increases (use: a thermostat).
- ☐ LDR: resistance decreases as light increases (use: lights that switch on when it gets dark).
- ☐ Sketch and explain the I–V graphs of a resistor, a filament lamp and a diode.
- ☐ Find resistance from a graph: $R = V \div I$ at one point.
- ☐ Describe the circuit that measures a component's resistance, using an ammeter and a voltmeter.

4. Series and parallel

- ☐ Series: same current through each component; supply p.d. is shared; $R_{\text{total}} = R_1 + R_2$.
- ☐ Parallel: same p.d. across each branch; branch currents add up to the total; total resistance is less than the smallest resistor.

- ☐ Explain why adding resistors in series increases total resistance, and in parallel decreases it.
- ☐ Calculate current, p.d. and resistance in series circuits.

5. Mains electricity

- ☐ Direct p.d. stays in one direction; alternating p.d. keeps reversing.
- ☐ UK mains is ac, about **230 V** at **50 Hz**.
- ☐ Three-core cable: live (brown), neutral (blue), earth (green and yellow), and the job of each.
- ☐ Live is about 230 V; neutral is close to 0 V; earth is 0 V and carries current only in a fault.
- ☐ Explain why the live wire is dangerous even with the switch off, and why any live-to-earth connection is dangerous.

6. Power and energy

- ☐ Use $P = VI$ and $P = I^2 R$ (W, V, A, Ω).
- ☐ Use $E = Pt$ and $E = QV$ (J, W, s, C, V).
- ☐ Energy transferred depends on the power and on how long the appliance is on.
- ☐ Describe how appliances transfer energy, for example to a motor's kinetic energy or to heating.
- ☐ Link an appliance's power rating to the energy it transfers.

7. The National Grid

- ☐ The National Grid is cables and transformers linking power stations to consumers.
- ☐ Step-up transformers raise the p.d. for the cables; step-down transformers lower it for homes.
- ☐ Explain why the Grid is efficient: higher p.d., smaller current, less heating of the cables.
- ☐ **Higher tier:** use $V_p I_p = V_s I_s$ (given on the equation sheet).

8. Required practicals

- ☐ **RP15:** how the length of a wire affects its resistance; resistors in series and in parallel.
- ☐ **RP16:** the I–V characteristics of a resistor, a filament lamp and a diode.

Equations to learn by heart: $Q = It$ $V = IR$ $P = VI$ $P = I^2 R$ $E = Pt$ $E = QV$

Given on the equation sheet (Higher tier): $V_p I_p = V_s I_s$

Practise with the P2 review packet on your class page.